Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 8 October 2026 Time: Not recorded
Group members present	Rohan Mathew, Jianhao Zhang, Xuan Zhang, Ria Mishra, Xingcan Wang (Alex). Supervisor: Saumya.
Apologies	None

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Implement the allocated five step shared pipeline.	All members	Progress presented
Prepare draft report and presentation slides for review.	All members	This meeting

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Fair comparison and statistical evidence	Use consistent preprocessing and justify sampling choices and synthetic assumptions. Saumya recommended Friedman and Wilcoxon signed-rank tests, p-values and critical difference diagrams where appropriate.
Data and conventional analysis	Jianhao presented a reproducible inventory and standardized data. Xuan presented LAOS metrics and strain-amplitude trends; analysis counts still needed reconciliation before the report was frozen.
Step response analysis	Rohan presented step size/direction and transient/stable comparisons. Undershoot and initial slope were informative, whereas a preplanned direction rule achieved only about 55%; this failure should be reported.
Real and synthetic evidence	Real data addressed relevance and instrument agreement; synthetic tests addressed known events. Four valid synthetic step runs gave preliminary CWT/DWT AUCs around 0.97/0.96, but low-resolution real step data limited wavelet support.
Practical recommendations	Alex proposed a compact set spanning conventional LAOS metrics, Fourier summaries and step response features. FFT was supported for periodic signals; wavelet potential required adequate temporal resolution and further validation.
Presentation requirements	Keep within 15 minutes, include all members with balanced contributions and practise rather than read a script. Explain the client, goals, accessible methods and key conclusions; the full result set belongs in the report.
Slide revisions and schedule	Create one connected story, reduce text, sharpen figures and define technical concepts. Use labels/reveals or a pointer to guide attention. Another rehearsal was agreed for Wednesday at 10:30 am; subsequent time would focus on report feedback.

Copyright © The University of Melbourne

3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Add appropriate statistical comparisons and report significance and limitations.	Analysis owners	Before analysis finalization
Reconcile measurement counts and document exclusions consistently.	Jianhao and Xuan	Before report is frozen
Finalize analysis and focus subsequent work on report and slides.	All members	Weekend following meeting
Revise slide flow, reduce text, improve graph clarity and simplify narration.	All members	Before next rehearsal
Share revised slides with Saumya for feedback.	All members	Before next rehearsal
Rehearse the full presentation with balanced contributions within 15 minutes.	All members	Next meeting

4 Next meeting

Location	Not recorded
Date & Time	Wednesday 14 October 2026, 10:30 am (Wednesday date inferred from the meeting date)
Goals	Rehearse the final presentation and review clarity, flow, timing and recommendations.

Copyright © The University of Melbourne